Golden Ratio Quantisation of Gravitational Wave Peak Frequencies and Dark Matter Mass from the φ-Lattice

Marvin L. Gentry
Independent Researcher, Seattle, WA
drmlgentry@protonmail.com ORCID: 0009-0006-4550-2663

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Abstract

The gravitational wave-induced freeze-in mechanism of Maleknejad and Kopp [1] shows that a stochastic gravitational wave (GW) background in the early Universe can produce massless Weyl fermions which, if they later acquire mass, constitute dark matter. We observe that this mechanism provides a natural cosmological realisation of the golden ratio lattice $m = m_e\varphi^{q/4}$ that governs the Standard Model fermion mass spectrum in the Hyperbolic Flavor Geometry (HFG) framework. We conjecture that the GW peak frequency is quantised in units of $\log\varphi$: $q_{\rm peak}/\mathcal{H}_* = \varphi^k$ for integer k, and that the produced dark matter mass satisfies $M_{\rm DM} = m_e\varphi^{q/4}$. The relic density constraint then relates k and q, yielding falsifiable predictions for future GW detectors (Einstein Telescope, Cosmic Explorer) and direct dark matter searches.

1 Introduction

The Hyperbolic Flavor Geometry (HFG) framework [2, 3] establishes that the fermion mass spectrum obeys

$$m = m_e \cdot \varphi^{q/4}, \qquad q \in \tfrac{1}{4}\mathbb{Z}, \tag{1}$$

where $\varphi = (1+\sqrt{5})/2$ and m_e is the electron mass. The Lucas sequence $L_k = \varphi^k + \varphi^{-k}$ governs the covering tower primes of the flavor manifolds, and $\log\varphi$ is the fundamental geodesic length unit.

Maleknejad and Kopp [1] recently showed that a stochastic GW background breaks the conformal symmetry of massless Weyl fermions, producing them via a 1-loop graviton–fermion vertex. The produced fermion energy density scales as [1]

$$\Omega_{\rm DM,0} \approx 0.36\,\mathcal{C}\left(\frac{g_*}{106.75}\right)^{4/3}\left(\frac{\Omega_{\rm peak}}{10^{-6}}\right)\left(\frac{M}{T_*}\right)\left(\frac{q_{\rm peak}/\mathcal{H}_*}{100}\right)^4\left(\frac{T_*}{3\times 10^{11}\,\mathrm{GeV}}\right)^5, \tag{2}$$

where $\mathcal{C} = \mathcal{O}(1)$ depends on the GW spectral index and coherence, T_* is the production temperature, $\mathcal{H}_*$ is the comoving Hubble scale at T_*, and $\Omega_{\rm peak}$ is the peak GW spectral energy density. Here we point out that equation (2) can accommodate the φ-lattice in a natural and falsifiable way.

2 φ-lattice quantisation

2.1 Peak frequency quantisation

The natural scale for any logarithmically structured spectrum in the HFG framework is $\log\varphi$, the regulator of $\mathbb{Q}(\sqrt{5})$ and the fundamental geodesic unit of the flavor manifolds [2]. We conjecture:

Conjecture 1 (φ-quantised GW spectrum)**.** *The peak frequency of a primordial stochastic GW background satisfies*

$$\frac{q_{\text{peak}}}{\mathcal{H}_*} = \varphi^k, \qquad k \in \mathbb{Z}_{\geq 0},$$

i.e., $\log(q_{\text{peak}}/\mathcal{H}_*) = k\log\varphi$.

Typical GW sources give $q_{\text{peak}}/\mathcal{H}_* \sim 10$–$10^3$ for first-order phase transitions [4], corresponding to $k \sim 5$–15 (since $\varphi^{10} \approx 123$, $\varphi^{15} \approx 1364$).

2.2 Dark matter mass on the φ-lattice

If the produced fermions acquire mass via a Higgs mechanism, the HFG framework predicts $M_{\text{DM}} = m_e\varphi^{q/4}$ for some $q \in \frac{1}{4}\mathbb{Z}$ not in the visible sector Lucas set [3]. Representative dark sector candidates from non-Lucas covering tower primes are listed in Table 1.

Table 1: Dark sector mass candidates from non-Lucas covering primes.

Prime p	$q = p$	$M = m_e\varphi^{p/4}$	Note
13	13	1.85 MeV	sterile neutrino range
19	19	6.4 MeV	light dark matter
31	31	39 MeV	dark sector mediator
41	41	580 MeV	sub-GeV dark matter
43	43	940 MeV	near proton mass

2.3 Consistency relation and falsifiability

Substituting Conjecture 1 and $M = m_e\varphi^{q/4}$ into equation (2) with $\Omega_{\text{DM},0} = 0.27$ gives a constraint on the integers k and q:

$$\frac{m_e}{T_*}\,\varphi^{4k+q/4} \approx \frac{0.75}{\mathcal{C}}\left(\frac{106.75}{g_*}\right)^{4/3}\left(\frac{10^{-6}}{\Omega_{\text{peak}}}\right)\left(\frac{T_*}{3\times 10^{11}\,\text{GeV}}\right)^{-5}. \tag{3}$$

For each choice of $(T_*, \Omega_{\text{peak}}, \mathcal{C})$, equation (3) selects a discrete set of integer pairs (k, q).

Proposition 2 (Falsifiable prediction)**.** *For a GW background detected with peak frequency f_{peak} (in Hz today), the ratio $\log(f_{\text{peak}}/H_0)/\log\varphi$ should be close to an integer k. The dark matter particle produced by this mechanism must have mass $M_{\text{DM}}/m_e = \varphi^{q/4}$ for some q not in the Lucas set. Both conditions can be tested independently.*

Remark 3 (The 17 MeV anomaly)**.** The claimed 17 MeV dark photon anomaly implies $q/4 \approx 7.28$, i.e., $q \approx 29.1 \approx L_7$ (a Lucas prime). The HFG framework therefore predicts this anomaly is *visible sector*, not dark sector — independently of the GW mechanism.

3 Discussion

The stochastic GW background acts as a "cosmic ruler" imprinting the φ-lattice on the dark matter mass through the production mechanism. The lattice would then appear in three observationally distinct places:

1. The GW peak frequency ratio $q_{\text{peak}}/\mathcal{H}_* = \varphi^k$ (detectable by Einstein Telescope [5] or Cosmic Explorer [6] in the kHz–GHz band).
2. The dark matter mass $M_{\text{DM}} = m_e\varphi^{q/4}$ (testable by direct detection or collider experiments).
3. The visible sector fermion masses already confirmed by the HFG framework [2].

The Maleknejad–Kopp mechanism provides the first proposed *physical mechanism* by which the φ-lattice could be imprinted cosmologically, as opposed to being postulated as a symmetry of the low-energy effective theory. If the GW background itself arises from a phase transition governed by the same arithmetic hyperbolic geometry, the quantisation of q_{peak} would follow naturally from the $\log\phi$ geodesic structure of the flavor manifolds.

4 Conclusion

The gravitational wave-induced freeze-in mechanism provides a cosmological arena in which the φ-lattice of the HFG framework can be tested. The conjecture $q_{\text{peak}}/\mathcal{H}_* = \varphi^k$ and the dark matter mass constraint $M_{\text{DM}} = m_e\varphi^{q/4}$ together with the relic density equation (2) yield a discrete, falsifiable prediction. Confirmation by future GW detectors would establish the φ-lattice as a fundamental structure of nature beyond the Standard Model flavor sector.

Data Availability

No new data were generated.

Conflict of Interest

The author declares no conflicts of interest.

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